TASK 1 Basic Conditional Satement and Looping

(A) Given 5 numbers, how many are even or odd

Aim: To design an algorithm and java program that takes 5 numbers as input and determines how many of them are even and how many are odd

Algorithm

(1) Start
(2) Initialize two conters
(3) Read 5 numbers from the user.
(4) for each number
* If the number%2 ==0 → increment even count
* else → increment odd count
(5) Display even count and odd count.
(6) Stop.

Program

```
import java.util.Scanner;
public class EvenOddCounter{
  public static void main(string[] args){
      Scanner sc =newScanner (system.in);
      int even count =0 , odd count =0 ;
      int [] numbers =new int[5];
   system.out.println(" Enter 5 numbers :");
   for(int i =0; i<5 ;i++){
      numbers[i] = sc.nextInt();
   if (numbers[i] %2 ==0){
    }else{
       odd count ++;
     }
     }
 }
  System.out.println(" Total Even numbers !" +even count);
  system.out.println("Total odd Number !"+ odd count);
  sc.close()
 }
}
```

Result: The given program was successfully verified and executed.

Input:

enter 5 numbers:

12, 15, 9, 4, 3

Output:

Total Even Number : 2

Total odd Numbers : 3

(B) Sum of last digit of two given numbers.

Aim: To write a program that finds the sum of ~~the~~ last digit of two given numbers.

Algorithm

(1) Start
(2) Input two nos, say a and b
(3) Find the last digit of 'a' using a%10.
(4) Find the last digit of 'b' using b%10.
(5) Add these two last digits
(6) output the result.
(7) Stop.

Program

```
import java.util.Scanner;
Public class LastDigitSum {
public Static void main (String[] args){
    Scanner sc = new Scanner (System.in);
    System.out.println(" Enter first number:");
    int a =sc.nextInt();
    System.out.println(" Enter second number:");
    int b = sc.nextInt();
    int lastDigit A = a%10;
    int lastDigit B=b%10;
    int sum =lastDigitA + lastDigitB;
    system.out.Println(" Sum of last digits = " + Sum)
    Sc.close();
}
```

Result: The given Program was successfully verified and executed.

Input

Enter first Number : 22
Enter second Number : 28

Output

Sum of last degits = 10

(C) To check whether a given number is prime

Aim: To design and implement a java program that determines whether a user-inputted integer is a prime number.

Algorithm:

(1) Start
(2) Initialize an integer variable flag initialized to.
(3) Read an integer number from the user.
(4) If num <= 1 , set flag = 0
(5) If n > 1 , start a loop with a counter i begin at 2
(6) if n % i == 0
(7) If flag == 1 display is prime
(8) stop.

Program:

```
import java.util Scanner;
class Isprime {
    public static void main (string[] args){
    Scanner sc = new Scanner (System.in);
    System.out.println ("Enter a number :");
    int num = sc.nextInt();
    int flag = 1;
    if (num <= 1){
        flag = 0;
    } else {
        for (int i = 2 ; i <= num/2 ; i++){
            if (num % i == 0){
                flag = 0;
                break;
            }
        }
    }
    if (flag == 1){
        System.out.println (`Prime number`);
    } else {
        System.out.println (` Not a prime number");
    }
    }
}
```

Result: The given program was successfully verified and executed.

Input:

enter a number : 66

Output:

Not a prime number.

(D) Factorial of a number

<u>Aim</u> : To calculate the factorial of a given number which Positive integer using an iterative approach

$$n! = n(n-1)(n-2)\ldots$$

<u>Algorithm</u> :

(1) Start
(2) Read an integer num from the user.
(3) set fact = 1
(4) initialize i = 1 to i<= num
-> multiply fact by i and get the fact value with new value
(5) After the loops ends, fact contains the factorial of num print the ~~r~~ result.
(6) stop.

<u>Program</u> :

```
import java.util.Scanner;
class factorial {
   public static void main (string[] args){
   Scanner sc  = new Scanner (system.in);
   System.out.print("enter a number:");
    int num = sc.nextInt();
    int fact = 1;
    for (int i=1; i<=num, i++){
       fact = fact * i;
    }
    system.out.println(" factorial of "+num+ "is: "+ fact);
}
}
```

<u>Result</u> : Hence, the above java code was successfull and executed

Input

enter a number : 42

Output

factorial of 42 is : 0

(E) Nth fibonacci

Aim: To develop a java program that reads an integer N from the user and prints the Nth fibonacci number using an iterative approach.

Algorithm:

(1) Start
(2) The user to enter the value of n.
(3) Initialize variables a=0, b=1, c
(4.) If n=1 then output a.
if n=2 then output b.
(5) Repeat from i=3 to n, c=a+b, a=b, b=c
(6) End.

Program:

```
import java.util.Scanner;
class Nth fibonacci{
public static void main (String[] args){
Scanner sc = new Scanner (System.in);
System.out.print(" Enter value of N:");
int n = sc.nextInt();
int a=0, b=1, c;
if(n==1){
    System.out.println("Nth fibonacci number is:" +a);
}else if (n==2){
    System.out.println(" Nth fibonacci number is :" +b);
}else {
    for (int i=3 , i<=n; i++){
        c=a+b;
        a=b;
        b=c
    }
    System.out.println(" Nth fibonacci number is:" +b);
  }
}
```

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PERFORMANCE (5)	5
RESULT AND ANALYSIS (3)	3
VIVA VOCE (3)	3
RECORD (4)	4
TOTAL (15)	15
SIGN WITH DATE	03/02/26

Result: Hence the java program was successfully executed

Input

Enter a number n = 56

Output

nth fibonacci numer is 225851433717